Name: ____________________________

Starter quiz

Converting fractions to recurring decimals

1 Terminating decimals have a ____________________ number of digits after the decimal point. Fill in the blank

2 Without using a calculator, work out the decimal value of $\frac{13}{20}$. ____________________ Fill in the blank

3 Select all the fractions that can be written as terminating decimals. Tick **3** correct answers

- ☐ $\frac{11}{15}$
- ☐ $\frac{11}{4}$
- ☐ $\frac{11}{8}$
- ☐ $\frac{11}{50}$
- ☐ $\frac{11}{30}$

4 Aisha uses her calculator to convert 0.668 to a fraction. She gets an answer of $\frac{167}{\square}$. What number should go in the square? ____________________ Fill in the blank

5 Match each fraction to its terminating decimal. You can use a calculator for this question. Write the correct letter in each box

a	$\frac{73}{160}$
b	$\frac{57}{125}$
c	$\frac{63}{140}$
d	$\frac{2281}{5000}$
e	$\frac{34}{85}$

	0.456
	0.45625
	0.4562
	0.4
	0.45

6 Sam says that $\frac{231}{330}$ is not equivalent to a terminating decimal. Without using a calculator, explain whether Sam is correct or not. Tick **1** correct answer

- ☐ Sam is correct as 330 has factors of 3 and 11
- ☐ Sam is correct as only fractions with a denominator of 10 terminate
- ☐ Sam is wrong as 330 is a multiple of 10
- ☐ Sam is wrong as the fraction simplifies to $\frac{7}{10}$ which terminates